

Modulation Signal Optimized Synchronous PWM

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The suppression of low-order harmonics is critical for the widely used large-radius machines, where the resulting vibration is amplified by zeroth-mode mechanical resonance, causing large acoustic noise. Although synchronous pulse-width modulation (PWM) is typically used to drive large machines, it inevitably introduces low-order harmonics. Optimized pulse patterns (OPPs) can eliminate arbitrary harmonics while minimizing phase-current distortions; however, it requires dedicated hardware for implementation. This paper first identifies a constraint on the switching instants of synchronous PWM and subsequently proposes a novel modulation method based on synchronous PWM. The proposed method achieves OPP-like optimization under this constraint without requiring dedicated hardware. In addition, torque and stator current control is implemented as in conventional synchronous PWM. The proposed method is validated through simulations and experiment. The results indicate that the proposed method almost eliminates the problematic harmonics caused by synchronous PWM while suppressing current distortion. The experimental results indicate that the elimination reduced the worst-case zeroth-mode vibration by half.

Keywords: synchronous PWM, optimized pulse pattern, vibration, acoustic noise, optimization

1. Introduction

The demand for saving energy and resources has encouraged broad use of interior permanent-magnet synchronous machines (IPMSMs) with reduced permanent-magnet as well as synchronous reluctance machines without permanent-magnets. These machines can generate reluctance torque utilizing the rotor saliency. However, spatial distribution of the inductance includes harmonics, which cause excitation force⁽¹⁾. Especially, large-radius machines, e.g., for traction applications, have natural frequency of mode zero in the audible range, which amplifies the vibration to cause large acoustic noise⁽²⁾⁽³⁾. For this reason, the spatial harmonics are generally designed to be sufficiently low.

Inverters for large machines widely use synchronous pulse width modulation (PWM) to supply stator current with low distortion exploiting limited switching frequency⁽⁴⁾. However, when the number of switching per fundamental periods decreases, low-order time harmonics are unavoidable. Particularly, $6n \pm 1$ th order harmonics excite zeroth-mode vibration⁽²⁾⁽³⁾.

To suppress acoustic noise, $6n \pm 1$ th order harmonics in stator current were reduced by control⁽⁵⁾⁽⁶⁾. Furthermore, stator current or flux linkage to compensate the vibration were determined beforehand by simulations or measurement

and applied to control⁽²⁾⁽³⁾⁽⁷⁾⁽⁸⁾. These methods are not applicable when the number of switching is low.

For the low number of switching, optimized pulse patterns (OPPs) were obtained by offline calculation to eliminate arbitrary harmonics⁽⁹⁾⁽¹⁰⁾. In addition, OPPs can form preferable distribution of harmonics and can minimize distortion of phase current. Model predictive pulse pattern control (MP³C) achieved feedback control using OPPs⁽¹¹⁾. However, OPPs and MP³C additionally require dedicated hardware like field-programmable gate arrays (FPGAs) for implementation. On the other hand, using synchronous PWM, the phase angle of a carrier signal against a modulation signal was adjusted so that time harmonics offset spatial harmonics⁽¹²⁾. However, this method cannot reduce time harmonics themselves, and the offsetting requires detailed characteristics of spatial harmonics of machines. In addition, the control requires precise information of the rotor position.

This paper with Ref. (13) proposes a novel modulation method based on synchronous PWM. Theoretically, the proposed method can eliminate the problematic harmonics while reducing the current distortion like OPPs but without the dedicated hardware. First, basic operation of synchronous PWM and OPPs is described, which reveals a constraint of synchronous PWM. Under this constraint, switching angles of synchronous PWM are optimized and then converted into modulation signals. After this procedure, the proposed method is named modulation signal optimized synchronous (MSOS-) PWM. Characteristics of the proposed method are first investigated in detail by simulations, and then verified by experiment. In addition, the impact on zeroth-mode vibration is evaluated by experiment. Common symbols are summarized in Table 1.

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Table 1. Nomenclature

f_c	Carrier frequency
f_s	Stator electrical frequency
N_p	the number of pulses (the ratio of f_c to f_s)
N_a	the number of adjustable switching angles
T	Period
α	Switching angles
θ	Phase, position

Fig. 1. Operating waveforms of synchronous PWM with $N_p = 9$ (Type B)

2. Synchronous PWM

The carrier frequency of synchronous PWM is integer multiples of stator electrical frequency. The frequency ratio, i.e., the number of pulses, is generally odd numbers and multiples of the number of phases. The negative or positive zero crossing of the carrier signal is ordinarily synchronized with zero degree of the sinusoidal modulation signal⁽⁴⁾, and the phase voltage starts with positive or negative values, respectively. The voltage waveform in the former and latter case is called type A and type B, respectively, in this paper.

Operating waveforms of synchronous PWM are shown in Fig. 1 for $N_p = 9$. When the number of pulses is an odd number and the abovementioned carrier signals are used, the phase voltage have half-cycle symmetry $v_{ph}(t+T/2) = -v_{ph}(t)$ and quarter-cycle symmetry $v_{ph}(T/2 - t) = v_{ph}(t)$.

Major harmonic frequency components caused by synchronous PWM are $f_c \pm 2f_s$ and $2f_c \pm f_s$ ⁽⁹⁾. Hence, the harmonic frequency to excite zeroth-mode vibration f_{M0} is $(N_p \pm 3)f_s$ and $2N_p f_s$ ⁽¹²⁾. Typical pattern of f_{M0} is shown in Fig. 2. In the figure, the number of pulses decreases with stator electrical frequency to limit the maximum carrier frequency. The harmonics $6f_s$ with $N_p = 9$ and $12f_s$ with $N_p = 15$ often overlap with natural frequency of mode zero for large-radius machines, and thus cause large vibration and acoustic noise.

3. Optimized Pulse Patterns

OPPs possess optimized harmonic distribution of phase voltage, and are obtained by adjusting switching angles for the optimization. OPPs starting with positive and negative phase voltage at zero degree of sinusoidal phase voltage is called type A and type B, respectively⁽⁹⁾.

An example of the phase voltage waveform is shown in Fig. 3. The phase voltage have half-cycle symmetry $v_{ph}(t+T/2) = -v_{ph}(t)$ and quarter-cycle symmetry $v_{ph}(T/2 - t) =$

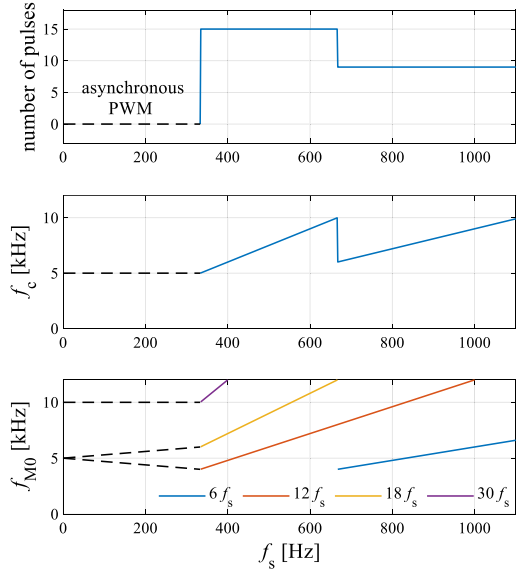


Fig. 2. Typical pattern of harmonic frequency to excite zeroth-mode vibration for stator electrical frequency

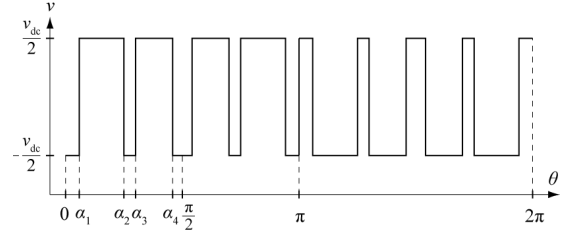


Fig. 3. The phase voltage waveform of OPPs (type B)

$v_{ph}(t)$. The former symmetry eliminates even-order harmonics and the latter cosinusoidal harmonics. Therefore, only odd-order sinusoidal harmonics remain in the phase voltage. In addition, $3n$ -order harmonics are cancelled in three-phase systems.

Utilizing the abovementioned symmetry, the phase voltage v_{ph} and the amplitude of k -order harmonics b_k are expressed by the following formula⁽⁹⁾.

$$v_{ph}(\theta) = \sum_{k=1}^{\infty} b_k \sin(k\theta) \dots\dots\dots (1)$$

$$b_k = \frac{4}{\pi k} \frac{v_{dc}}{2} \left\{ 1 + 2 \sum_{i=1}^{N_a} (-1)^i \cos(k\alpha_i) \right\} \dots\dots\dots (2)$$

By adjusting the switching angles based on these equations, the harmonic components can be controlled at preferable values. The relationship between the number of pulses and adjustable switching angles is expressed by

$$N_p = 2N_a + 1 \dots\dots\dots (3)$$

As for the optimization problem for OPPs, the fundamental component is controlled by the following nonlinear constraint

$$b_1 = \frac{v_{dc}}{2} M, \dots\dots\dots (4)$$

where v_{dc} is the dc-link voltage and M the modulation ratio. The distortion of phase current is minimized while optimizing distribution of the harmonic components by the following objective function

$$f_{obj} = \sum_{j=1}^8 \left\{ \left(w_{6j-1} \frac{b_{6j-1}}{6j-1} \right)^2 + \left(w_{6j+1} \frac{b_{6j+1}}{6j+1} \right)^2 \right\}, \dots\dots (5)$$

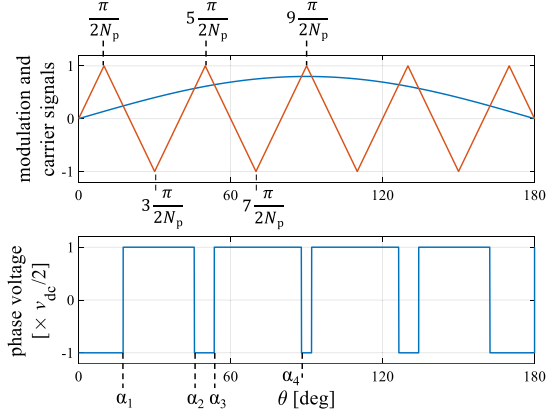


Fig. 4. The constraint on the range of switching angles of synchronous PWM ($N_p = 9$)

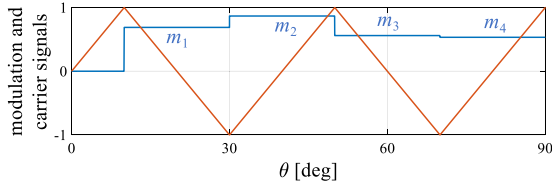


Fig. 5. Converted modulation signals to recreate optimized switching angles ($N_p = 9$)

where w_k is the weighting factor for the k -order harmonic and the harmonics up to 49th orders are considered. To evaluate the current distortion using phase voltage in simulations and experiment, this paper uses $W_{THD0}^{(9)}$.

$$W_{THD0} = \frac{\sqrt{\sum_{k=2}^{\infty} \left(\frac{V_k}{k}\right)^2}}{V_{1,M=1}} \dots \dots \dots (6)$$

In (6), $V_{1,M=1}$ is the amplitude of fundamental phase voltage with $M = 1$ and V_k the amplitude of the k -order harmonic.

In this paper, the harmonic orders 6 ± 1 for $N_p = 9$ and 12 ± 1 for $N_p = 15$ are reduced. Hence, w_5 and w_7 are set as large as at 10 for $N_p = 9$ while keeping other weighting factors 1. For $N_p = 15$, w_{11} and w_{13} are set at 10 similarly. To solve the optimization problem, a function: *fmincon* in MATLAB is used⁽¹⁴⁾.

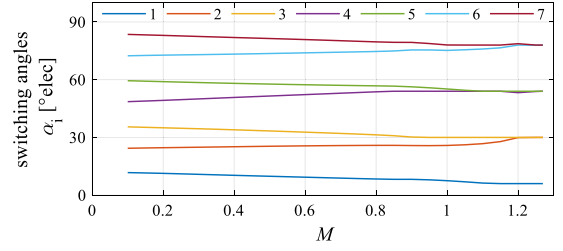
4. Proposed Method

4.1 Implementation of OPPs based on Synchronous PWM The difference between synchronous PWM and OPPs is that switching angles of synchronous PWM are constrained by the carrier signal as shown in Fig. 4. The constraint is expressed by

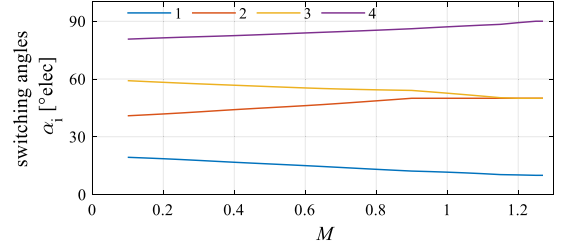
$$\frac{\pi}{2N_p}(2i-1) \leq \alpha_i \leq \frac{\pi}{2N_p}(2i+1) \quad (i = 1, \dots, N_a) \dots \dots \dots (7)$$

Therefore, OPPs obtained under the constraint (7) can be implemented based on synchronous PWM. For the implementation, optimized switching angles are converted into modulation signals.

For the converted modulation signals, the number of adjustable values is the same as the number of adjustable angles, and hence 4 for $N_p = 9$ as shown in Fig. 5. To recreate optimized switching angles simply and accurately,

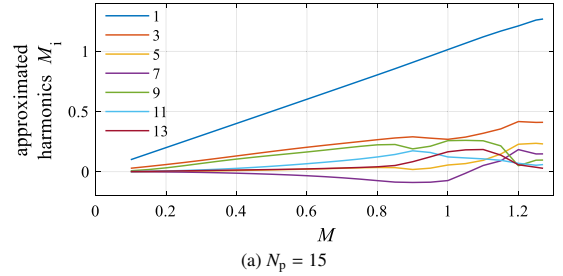


(a) $N_p = 15$

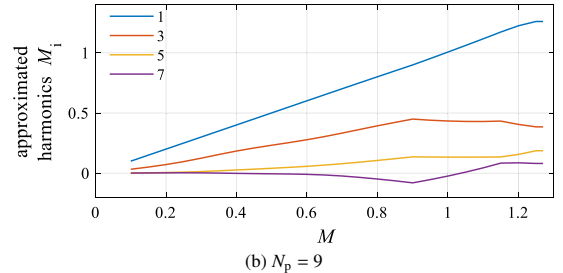


(b) $N_p = 9$

Fig. 6. Switching angles of MSOS-PWM



(a) $N_p = 15$



(b) $N_p = 9$

Fig. 7. Approximated harmonics of MSOS-PWM

the modulation signal is held constant during each half carrier period. The conversion is performed by

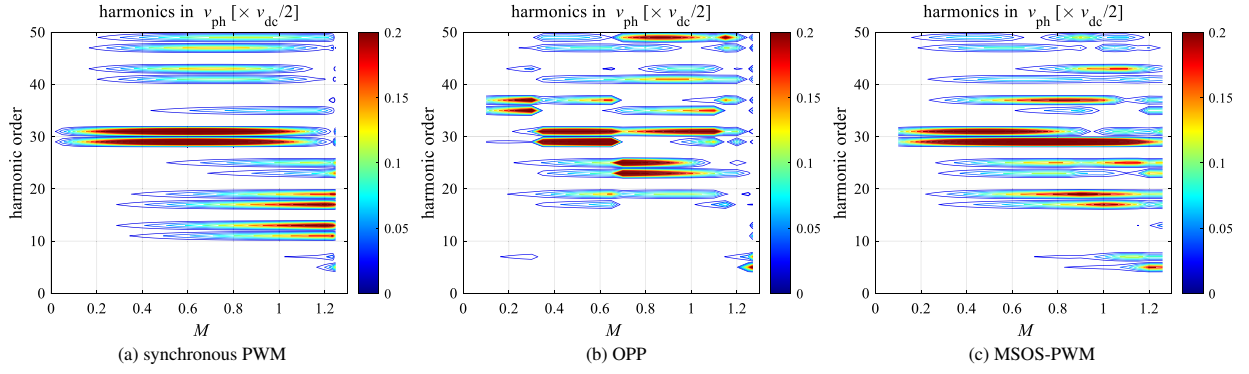
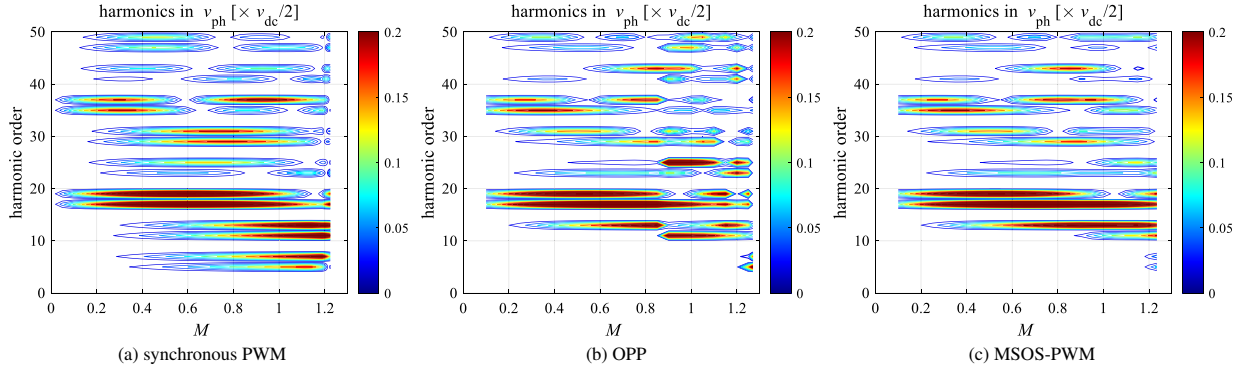
$$m_i = k_c(-1)^{i-1} \left(2 \frac{\alpha_i - \frac{\pi}{2N_p}(2i-1)}{\pi/N_p} - 1 \right) \quad (i = 1, \dots, N_a) \dots \dots \dots (8)$$

where the coefficient k_c is 1 for type A and -1 for type B.

With the converted modulation signals, OPPs are precisely reconstructed when the carrier signal is completely synchronized with the modulation signal. However, as the synchronization is not always maintained, modulation signals are approximated with functions.

The approximation requires N_a parameters, the number of which is the same as that of the adjustable angles. Considering the symmetric property of phase voltage, odd-order sinusoidal components are suitable for the function approximation. Therefore, modulation signals are approximated by

$$m_{ph} = \sum_{i=1}^{N_a} M_{2i-1} \sin\{(2i-1)\theta\} \quad (i = 1, \dots, N_a) \dots \dots \dots (9)$$

Fig. 8. Comparison of harmonic components of the three modulation methods with $N_p = 15$ Fig. 9. Comparison of harmonic components of the three modulation methods with $N_p = 9$

where N_α of modulation ratio of fundamental and harmonic components are employed in ascending order.

4.2 Characteristics of Harmonic Components

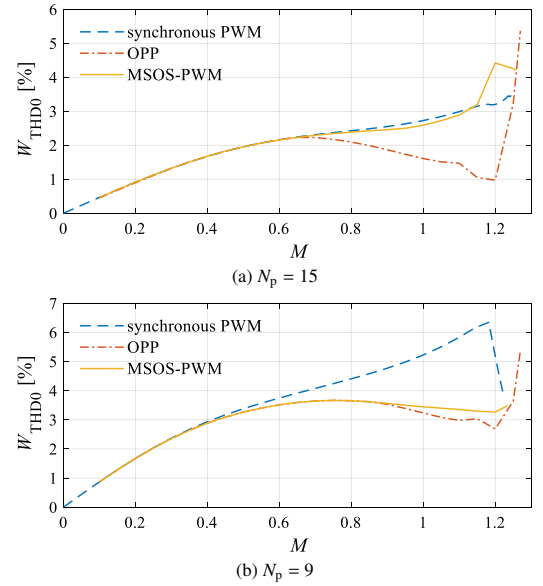
Characteristics of MSOS-PWM were calculated and compared with synchronous PWM and OPPs. The voltage waveform of type B was employed for all the modulation methods as it is favorable to reduce harmonic orders 6 ± 1 for $N_p = 9$ and 12 ± 1 for $N_p = 15$. Synchronous PWM used a method of third harmonic injection, i.e. pulse centering⁽¹⁵⁾, to expand a feasible range of the fundamental phase voltage.

Switching angles of MSOS-PWM are shown in Fig. 6. In the figure, the switching angles satisfy the constraint (7). The approximated harmonics are presented in Fig. 7.

Harmonic components of the three methods are compared in Fig. 8 for $N_p = 15$ and in Fig. 9 for $N_p = 9$. MSOS-PWM almost eliminates 12 ± 1 th order harmonics for $N_p = 15$ and 6 ± 1 th order harmonics for $N_p = 9$ similarly to OPPs. These harmonics are unavoidable for synchronous PWM.

The current distortion is compared between the three methods in Fig. 10. In the case $N_p = 15$, the current distortion of MSOS-PWM is almost the same as synchronous PWM and only OPPs can reduce the distortion. As for the case $N_p = 9$, the current distortion of MSOS-PWM is significantly reduced compared with synchronous PWM and is as low as OPPs.

4.3 Implementation MSOS-PWM is simply incorporated into control of the torque and stator current in a similar way to synchronous PWM. The relevant part of the control is shown in Fig. 11. Generally, current control calculates the stator voltage reference in the rotational d-q reference frame $\mathbf{v}_s^{dq*}$. This is transformed into the three-phase coordinate system using the rotor position θ_r , and then converted into the modulation signals m_u' , m_v' , m_w' , to which the approximated

Fig. 10. Comparison of the current distortion by W_{THD0} between the three modulation methods

harmonics of MSOS-PWM are added.

The modulation ratio M and the stator voltage phase based on sinusoidal signals $\theta_{s,\sin}$ are calculated from $\mathbf{v}_s^{dq*}$. In the calculation, the low-pass filters (LPFs) are installed to extract the fundamental components and hence the cut-off frequency is set 1/5 of response of the current control. The amplitude of approximated harmonics M_1 to M_i are obtained using the look-up tables $M_1(M)$ to $M_i(M)$, respectively.

5. Experimental Verification

Effectiveness of MSOS-PWM was verified, and the characteristics were compared with synchronous PWM by

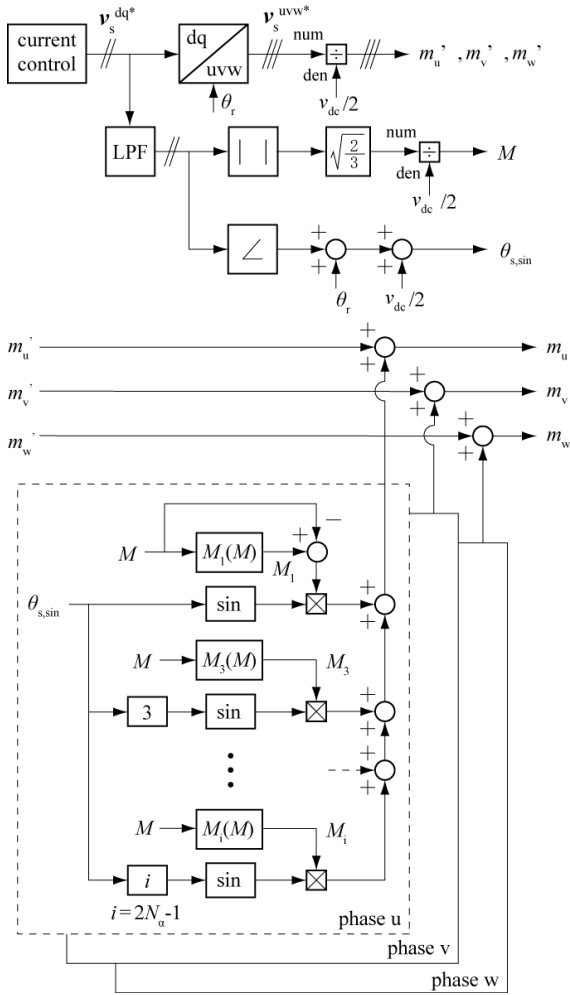


Fig. 11. Block diagram of MSOS-PWM incorporated into current control

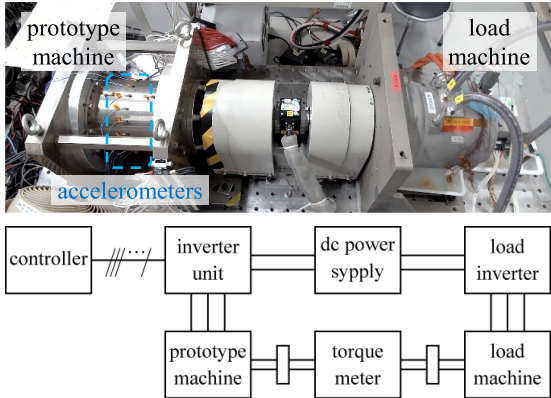


Fig. 12. The experimental apparatus and system

experiment. The experimental apparatus and system are shown in Fig.12. The prototype machine, specified in Table 2, is an IPMSM with distributed windings and was driven by the inverter unit with 5th generation IGBTs. Eighteen accelerometers were attached on the circumference of the prototype machine and zeroth-mode vibration was calculated from the in-phase components. As the controller, PE-Expert4 with a digital signal processor and a FPGA⁽¹⁶⁾ was used to implement synchronous and MSOS-PWM.

In the experiment, the machine speed was controlled by the load inverter, and the torque was regulated by the controller

Table 2. Specifications of the prototype machine

Maximum torque	30 Nm
Maximum phase current (rms)	100 A
Number of pole pairs	9

Table 3. Parameters of the inverter unit and control

Dc-link voltage	100 V
Crossover frequency of the current control	333 rad/s
Dead time	5 μ s

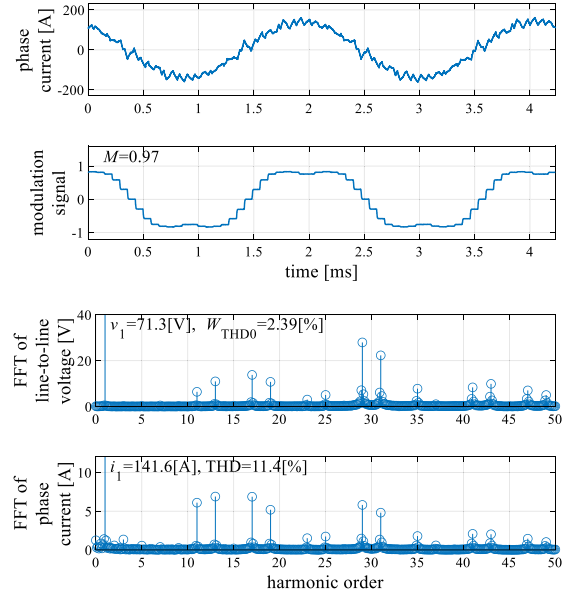


Fig. 13. Operating waveforms of synchronous PWM with $N_p = 15$ at $n_r = 3150$ r/min and $T^* = 30$ Nm

using the field-oriented control⁽¹⁵⁾. The torque reference was converted into the stator current reference through look-up tables, and then proportional-integral control was used. Parameters of the inverter unit and control was set as described in Table 3. The control was basically implemented at the top and bottom of the carrier signal for both methods. The dead-time compensation was not installed in this paper.

5.1 Steady States Operating waveforms with $N_p = 15$ at $n_r = 3150$ r/min and $T^* = 30$ Nm are shown in Fig. 13 for synchronous PWM and in Fig. 14 for MSOS-PWM. The results of frequency analysis (fast fourier transform, FFT) confirm that MSOS-PWM almost eliminated 12 ± 1 th order harmonics from the stator line-to-line voltage while increasing 6 ± 1 th order harmonics slightly. Consequently, 12 ± 1 th order harmonics in the stator phase current were significantly reduced. Note that there remained harmonic currents of the corresponding orders caused by spatial harmonics of the machine. However, distortion of the stator voltage and current was slightly increased by applying MSOS-PWM. The results also confirm that the approximated harmonics were added to the modulation signal of MSOS-PWM. Note that non-ideal characteristics of the inverter unit caused slight difference in the modulation ratio between the two methods although the difference in their control is only harmonics added to the modulation signal.

Operating waveforms with $N_p = 9$ at $T^* = 20\text{Nm}$ are shown in Fig.15 and Fig.16. The results demonstrate that MSOS-PWM significantly reduced the $6\pm 1\text{th}$ order

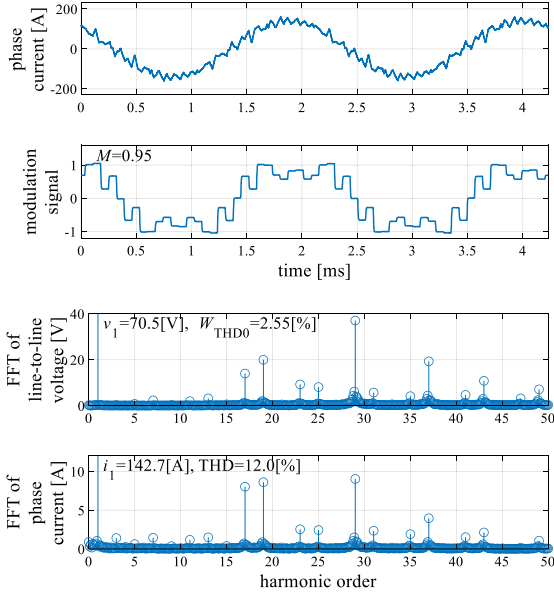


Fig. 14. Operating waveforms of MSOS-PWM with $N_p = 15$ at $n_r = 3150$ r/min and $T^* = 30$ Nm

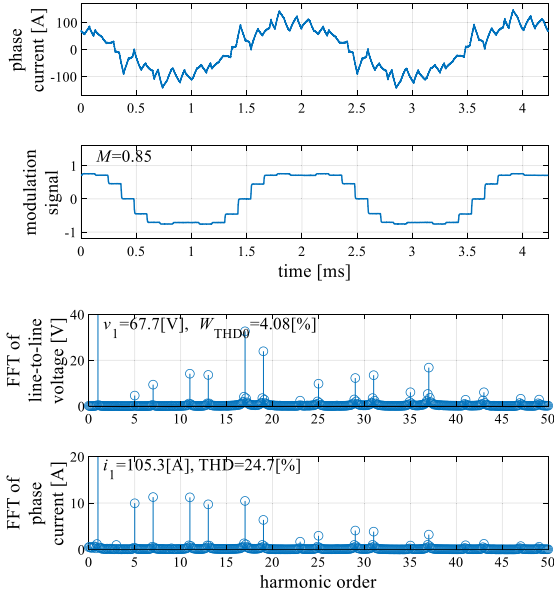


Fig. 15. Operating waveforms of synchronous PWM with $N_p = 9$ at $n_r = 3150$ r/min and $T^* = 20$ Nm

harmonics in a similar way. In addition, distortion of the stator voltage and current was substantially reduced.

In the experimental results, reduction of the distortion by MSOS-PWM was smaller than the calculation results with ideal conditions in Sec.4.2. To figure out the causes, influence of voltage errors of the inverter unit was investigated by simulations. As the voltage errors, dead time and voltage drops of the semiconductor devices were considered. For simulations, MATLAB/Simulink was used with a JMAG-RT model⁽¹⁷⁾ of the prototype machine and characteristics of the voltage drops were referred to the datasheet.

Simulation results of the distortion with and without the voltage errors are summarized in Fig. 17. From the results, synchronous PWM was robust against the voltage errors. On the other hand, distortion of MSOS-PWM increased with $N_p = 15$ and did not with $N_p = 9$ due to the voltage errors. Especially, dead time had strong influence. From

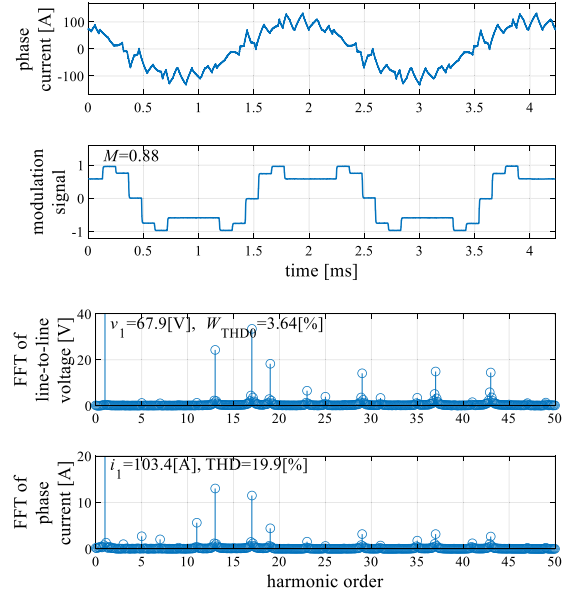


Fig. 16. Operating waveforms of MSOS-PWM with $N_p = 9$ at $n_r = 3150$ r/min and $T^* = 20$ Nm

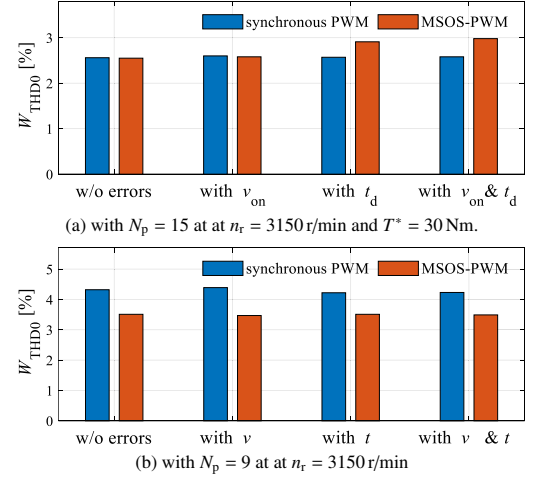


Fig. 17. Influence of voltage errors of the inverter unit to the distortion investigated by simulations

the modulation signals in operating waveforms (Fig. 13 to Fig. 16), occurrence of the deterioration depended on addition of the high-order harmonics to the modulation signal.

5.2 Transient Response

Transient response was evaluated with step response of the torque. In the experiment, the torque reference was changed by step functions from 5 Nm to 15 Nm at $t = 0$ s. The response was evaluated using the stator current as it is difficult to measure the torque with high response.

Experimental waveforms with $N_p = 15$ at $n_r = 3150$ r/min are shown in Fig. 18 for synchronous PWM and in Fig. 19 for MSOS-PWM. The results confirm that MSOS-PWM achieved almost the same transient response as synchronous PWM and the resulting time constant almost matched the designed value (3 ms). Small difference between the two methods is that the response of synchronous PWM was slightly faster. The reason is that MSOS-PWM added the approximated harmonics to modulation signals and consequently the margin for feedback control was reduced.

Note that ratio of ripples to fundamental components in the phase current was larger than that in Sec. 5.1 as the torque and

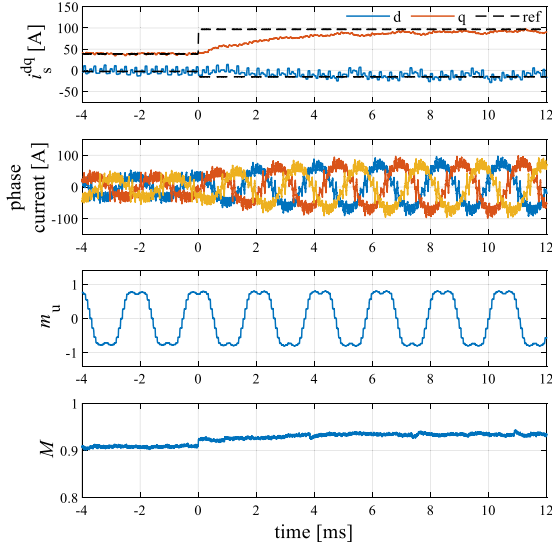


Fig. 18. Transient response of synchronous PWM with $N_p = 15$ for the torque step at $n_r = 3150$ r/min

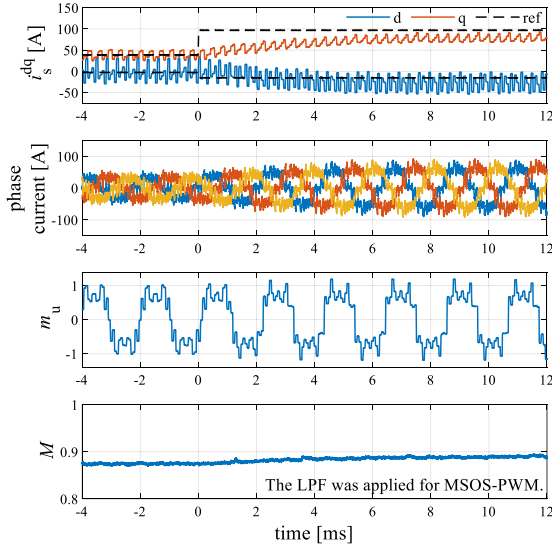


Fig. 19. Transient response of MSOS-PWM with $N_p = 15$ for the torque step at $n_r = 3150$ r/min

fundamental current was smaller. In addition, more ripples were reflected in the waveform of i_s^{dq} for MSOS-PWM. The reason is that i_s^{dq} was updated only at the top and bottom of the carrier signal and more harmonics appeared at these updating instants for MSOS-PWM. From the waveform of the phase current, amplitude of the ripples was almost the same for both methods.

Experimental results with $N_p = 9$ are shown in Fig. 20 and Fig. 21. In the results, both modulation methods achieved the designed response in a similar way.

5.3 Vibration The vibration of mode zero was evaluated at constant machine speed and torque. Target harmonic components were 12th order for $N_p = 15$ and 6th order for $N_p = 9$. Zeroth-mode vibration calculated from measured acceleration is shown in Fig. 22. The vibration in Fig. 22(a) was large because the frequency component was close to the natural frequency (5645 Hz).

The results show that the vibration varied depending on the torque with synchronous PWM, and stayed almost constant with MSOS-PWM. The reason is that the excitation force

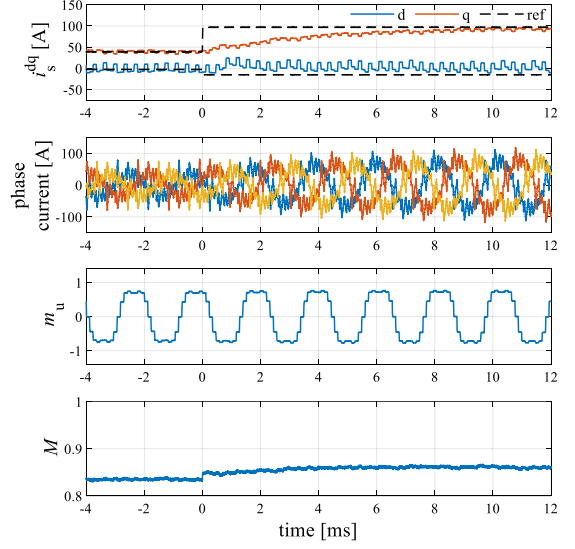


Fig. 20. Transient response of synchronous PWM with $N_p = 9$ for the torque step at $n_r = 3150$ r/min

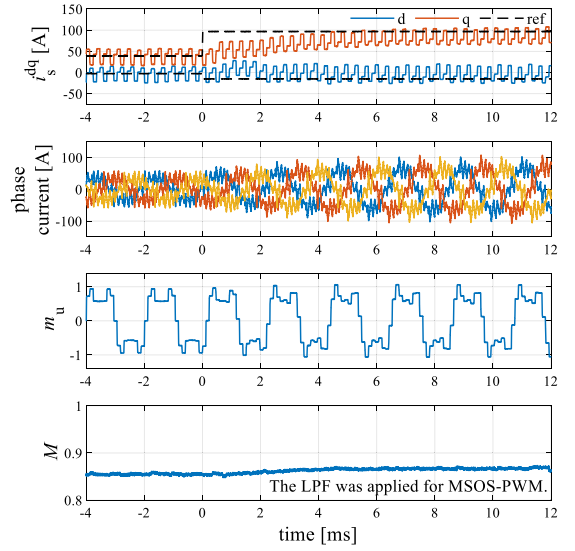
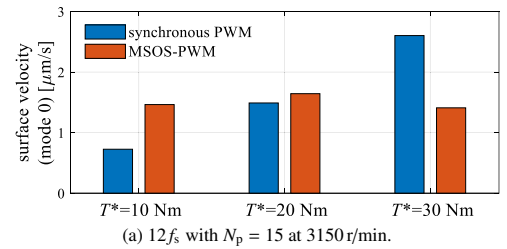
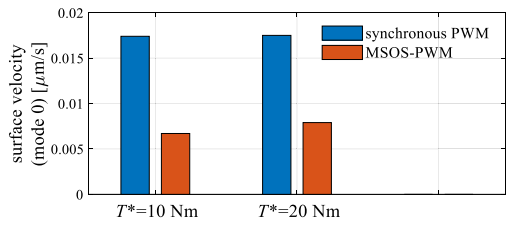


Fig. 21. Transient response of MSOS-PWM with $N_p = 9$ for the torque step at $n_r = 3150$ r/min



(a) $12f_s$ with $N_p = 15$ at 3150 r/min.



(b) $6f_s$ with $N_p = 9$ at 3150 r/min.

Fig. 22. Zeroth-mode vibration calculated from the measured acceleration

was synthesized from time harmonics of the inverter and spatial harmonics of the machine and MSOS-PWM almost eliminated time harmonics of the target orders. As a result, MSOS-PWM reduced the worst value of the vibration to half both for $N_p = 15$ and $N_p = 9$.

6. Conclusions

This paper proposed modulation signal optimized synchronous (MSOS-) PWM. Theoretically, the proposed method can eliminate harmonics of the problematic orders while reducing distortion of phase current like optimized pulse patterns but without the dedicated hardware. This superior performance is achieved by optimization of switching instants while satisfying the constraint due to implementation based on synchronous PWM. Control of the torque and stator current is simply implemented in a similar way to synchronous PWM.

Experimental verification demonstrated that MSOS-PWM almost eliminates time harmonics of the problematic orders, e.g., 6 ± 1 th orders for $N_p = 9$ and 12 ± 1 th orders for $N_p = 15$, even without dead-time compensation. This elimination reduced the worst value of zeroth-mode vibration to half which was synthesized from time harmonics of the inverter and spatial harmonics of the machine. The current distortion is substantially reduced in the case $N_p = 9$ and almost the same in the case $N_p = 15$ compared with synchronous PWM. However, the distortion slightly increases in the case $N_p = 15$ without dead-time compensation. As for feedback control, MSOS-PWM achieves almost the same response as synchronous PWM.

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